

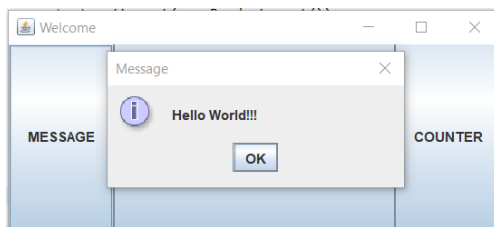
Practice Exercise GUI

Design and implement GUI program using java swing components JFrame and JPanel as follows:

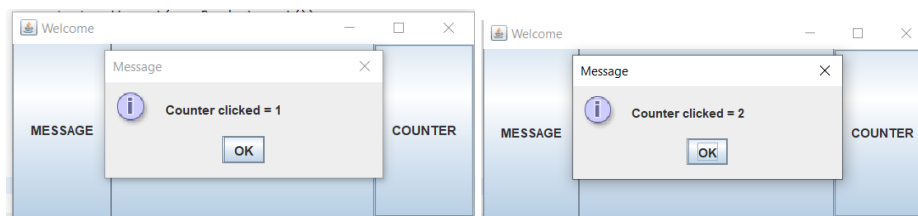
- GUI will have 3 buttons: COUNTER, MESSAGE, and EXIT



- When user clicks on MESSAGE, a Message Dialog will pop up with “Hello World!!!” string and OK Button. When user clicks on OK, the message dialog will close



- When user clicks on COUNTER, a Message Dialog will pop with the String “Counter clicked =” followed by the value of how many times the user clicked that button. So first time the Message will have 1, then 2, then 3 etc. When user clicks on OK, the message dialog will close



- When user clicks on EXIT, the program will terminate
- **NOTE:** You should use `JOptionPane.showMessageDialog` for the Message Dialog. You will need to pass the `JFrame` to the event handler to use as the first parameter in method.

Sample Solution

```
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;

public class Buttons {

    // counter event handler
    private static class CounterHandler implements ActionListener {

        private int clicks = 0;
        private JFrame window = null;

        // setup frame to be used in JOptionPane
        public CounterHandler(JFrame frame) {
            window = frame;
        }

        public void actionPerformed(ActionEvent e) {
            clicks++;
            String msg = "Counter clicked = " + clicks;
            JOptionPane.showMessageDialog(window, msg);
        }
    }

    // exit event handler
    private static class ExitHandler implements ActionListener {

        public void actionPerformed(ActionEvent e) {
            System.exit(0);
        }
    }

    // message event handler
    private static class MessageHandler implements ActionListener {

        private JFrame window = null;

        // setup frame to be used in JOptionPane
        public MessageHandler(JFrame frame) {
            window = frame;
        }

        public void actionPerformed(ActionEvent e) {
            JOptionPane.showMessageDialog(window, "Hello World!!!");
        }
    }

    public static void main(String[] args) {

        // create JButton instances
```

```

        JButton countButton = new JButton("COUNTER");
        JButton msgButton = new JButton("MESSAGE");
        JButton exitButton = new JButton("EXIT");

        // create frame
        JFrame window = new JFrame("Welcome");

        // add buttons to the panel
        JPanel content = new JPanel();
        content.setLayout(new BorderLayout());

        // create listeners and add to buttons
        CounterHandler countListener = new CounterHandler(window);
        countButton.addActionListener(countListener);
        ExitHandler exitListener = new ExitHandler();
        exitButton.addActionListener(exitListener);
        MessageHandler msgListener = new MessageHandler(window);
        msgButton.addActionListener(msgListener);

        // add buttons to panel widget to the pa
        content.add(countButton, BorderLayout.EAST);
        content.add(exitButton, BorderLayout.CENTER);
        content.add(msgButton, BorderLayout.WEST);

        // adds content we created above
        window.setContentPane(content);

        // setup size and location
        window.setSize(450,200); // sets size of the window
        window.setLocation(300,400); // sets x,y location where it should
display
        window.setVisible(true); // makes it visible
    }

```